

Maintenance Schedule

Frac Pack

EPA/CARB

TF12V4000C1

TF12V4000C2

TF12V4000C3

Application Group 4D

MSE50794/02E



Power. Passion. Partnership.

System designation	Engine model	kW/cyl.	Application group
TF12V4000C1	12V4000T95R	140 kW/cyl.	4D, Frac operation (continuous operation with low load)
TF12V4000C2	12V4000T95	155 kW/cyl.	4D, Frac operation (continuous operation with low load)
TF12V4000C3	12V4000T95L	162 kW/cyl.	4D, Frac operation (continuous operation with low load)

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

Information about the latest maintenance schedule is available at: <http://www.mtu-online.com>.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emission specifications. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-relevant components may only be serviced, replaced or repaired if components approved by MTU are used. Emission-relevant components are for instance control units, data records, sensors, injectors, exhaust flaps and all components of exhaust aftertreatment systems.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals and the scope of required maintenance activities are the result of operational experience and therefore represent recommendations. Severe operating and ambient conditions may require additional maintenance work and/or modification of the maintenance intervals.

Adherence to the specified maintenance intervals is essential to maintain product safety.

Maintenance activities are specified with intervals based on hours in operation and time limits. Whichever value is reached first shall apply.

The individual maintenance intervals are classified in accordance with the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

For change intervals for fluids and lubricants as well as for preservation specifications, refer to the relevant instructions of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available online at: <http://www.mtu-online.com>, and for power generation systems at: <http://www.mtuonsiteenergy.com>.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

1.2 Allocation matrix application group 4D

Load profile		12V/16V4000T95 / 12V4000T95x						
Load %	Time %	Load factor %	Load indicator %	121 to 140 kW/cyl.	141 to 163 kW/cyl.			
110 ₍₁₀₀₋₁₁₀₎	—	0 to 22	≤ 2	24,000	18,000	—	—	
100 ₍₉₀₋₁₀₀₎	1							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	5							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	15							
30 ₍₅₋₃₀₎	24							
Idle Speed ₍₀₋₅₎	55							
110 ₍₁₀₀₋₁₁₀₎	—	23 to 30	≤ 4	21,000	15,000	—	—	
100 ₍₉₀₋₁₀₀₎	2							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	10							
65 ₍₅₀₋₆₅₎	25							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	3							
Idle Speed ₍₀₋₅₎	60							
110 ₍₁₀₀₋₁₁₀₎	—	31 to 41	≤ 9	18,000	15,000	—	—	
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	25							
65 ₍₅₀₋₆₅₎	20							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle Speed ₍₀₋₅₎	50							
110 ₍₁₀₀₋₁₁₀₎	—	42 to 52	≤ 12	15,000	12,000	—	—	
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	5							
80 ₍₆₅₋₈₀₎	40							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	30							
Idle Speed ₍₀₋₅₎	20							

Table 2: Allocation matrix application group 4D

12V/16V4000T95 / 12V4000T95x							
Load profile		Load factor %	Load indicator %	121 to 140 kW/cyl.	141 to 163 kW/cyl.	-	-
Load %	Time %						
110 ₍₁₀₀₋₁₁₀₎	—	53 to 61	≤ 20	12,000	—	—	—
100 ₍₉₀₋₁₀₀₎	10						
90 ₍₈₀₋₉₀₎	15						
80 ₍₆₅₋₈₀₎	35						
65 ₍₅₀₋₆₅₎	—						
50 ₍₃₀₋₅₀₎	—						
30 ₍₅₋₃₀₎	30						
Idle Speed ₍₀₋₅₎	10						

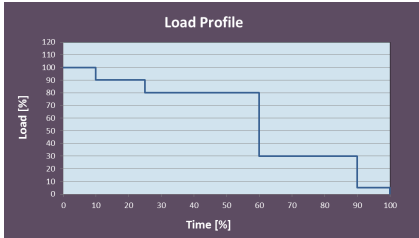


Table 3: Allocation matrix application group 4D

1.3 Maintenance schedule matrix 24,000 h

0 to 14,500 Operating hours

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
Engine																															
W0800	-																														
W0505	1 w																														
W1024	1 w																														
W1008	2 a																														
W1009	2 a																														
W0500	-	X																													
W0501	-	X																													
W0503	-	X																													
W0506	-	X																													
W0507	-	X																													
W0525	-	X																													
W8646	-	X																													
W8644	-	X																													
W8645	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1005	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1481	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1675	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a							X						X						X						X					
W1207	2 a			X				X						X						X						X					
W1551	2 a							X						X						X						X					
W1296	3 a							X						X						X						X					
W1817	18 a							X						X						X						X					
W1046	6 a									X								X								X					
W1525	6 a													X												X					
W1526	6 a													X												X					
W1610	18 a																		X												
W1577	18 a																		X												
W1689	18 a																		X												
W1006	18 a																		X												
W1636	18 a																		X												
W1011	18 a																		X												
W4184	18 a																									X					
W1818	18 a																	X													
W1250	6 a																									X					
W1251	6 a																									X					
W2012	18 a																									X					
W2001	18 a																									X					
W2006	18 a																									X					
W2007	18 a																									X					
W2009	18 a																									X					

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
W2010	18 a																									X					
W2062	18 a																									X					
W2070	18 a																									X					
W2110	18 a																									X					
W1051	18 a																									X					
W1735	18 a																									X					
W1038	18 a																									X					
W1043	18 a																														
W1134	18 a																														
W1175	18 a																														
W3000	18 a																														
W3001	18 a																														
W3002	18 a																														
W3003	18 a																														
W3004	18 a																														
W3005	18 a																														
W3007	18 a																														
W3018	18 a																														
W3021	18 a																														
W3103	18 a																														
W3108	18 a																														
W3177	18 a																														
W3182	18 a																														
W1124	18 a																														
W1901	18 a																														
W1819	18 a																														
Transmission (ZF)																															
X4032	-	X																													
X4033	-	X																													
X4034	-	X																													
X4035	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
X4023	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
X4037	-			X		X		X		X		X		X		X		X		X		X		X		X		X			
X4042	-																														
W4146	-																														
Air compressor																															
W1784	-																X														
w = weeks m = months a = years																															

15,000 to 24,000 Operating hours

Task	Limit	Operating hours [h]																												
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000										
Engine																														
W0800	-																													
w = weeks																														
m = months																														
a = years																														

Task	Limit	Operating hours [h]																											
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000									
W0505	1 w																												
W1024	1 w																												
W1008	2 a																												
W1009	2 a																												
W0500	-																												
W0501	-																												
W0503	-																												
W0506	-																												
W0507	-																												
W0525	-																												
W8646	-																												
W8644	-																												
W8645	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X									
W1005	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X									
W1001	2 a	X		X		X		X		X		X		X		X		X		X									
W1481	2 a	X		X		X		X		X		X		X		X		X		X									
W1675	2 a	X		X		X		X		X		X		X		X		X		X									
W1241	2 a	X		X		X		X		X		X		X		X		X		X									
W1575	1 a	X						X						X						X									
W1207	2 a	X						X						X						X									
W1551	2 a	X						X						X						X									
W1296	3 a	X						X						X						X									
W1817	18 a	X						X						X						X									
W1046	6 a			X								X								X									
W1525	6 a							X												X									
W1526	6 a							X												X									
W1610	18 a			X																X									
W1577	18 a			X																X									
W1689	18 a			X																X									
W1006	18 a			X																X									
W1636	18 a			X																X									
W1011	18 a			X																X									
W4184	18 a																			X									
W1818	18 a			X																X									
W1250	6 a																			X									
W1251	6 a																			X									
W2012	18 a																			X									
W2001	18 a																			X									
W2006	18 a																			X									
W2007	18 a																			X									
W2009	18 a																			X									
W2010	18 a																			X									
W2062	18 a																			X									
W2070	18 a																			X									
W2110	18 a																			X									
W1051	18 a																			X									
W1735	18 a																			X									

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																							
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000					
W1038	18 a																			X					
W1043	18 a																			X					
W1134	18 a																			X					
W1175	18 a																			X					
W3000	18 a																			X					
W3001	18 a																			X					
W3002	18 a																			X					
W3003	18 a																			X					
W3004	18 a																			X					
W3005	18 a																			X					
W3007	18 a						X					X								X					
W3018	18 a																			X					
W3021	18 a																			X					
W3103	18 a																			X					
W3108	18 a																			X					
W3177	18 a																			X					
W3182	18 a																			X					
W1124	18 a																			X					
W1901	18 a																			X					
W1819	18 a																			X					
Transmission (ZF)																									
X4032	-																								
X4033	-																								
X4034	-																								
X4035	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
X4023	1 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
X4037	-	X		X		X		X		X		X		X		X		X		X					
X4042	-																			X					
W4146	-																			X					
Air compressor																									
W1784	-			X																X					
w = weeks m = months a = years																									

1.4 Maintenance tasks 24,000 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL1	-	1 w	Engine operation	Check relief bores of coolant pump(s).		W0505
QL1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).	X	W1024
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL1	-	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.		W1009
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.		W0503
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Drain off water and contamination from fuel prefilter.		W0507
				Inspect battery-charging generator for contamination and clean if necessary.	X	W0525
QL1	Daily	-	Exhaust silencer	Check condensate drain of silencer.		W8646
QL1	Daily	-	Back Pack securing screws	Check for damage (visual inspection).		W8644
QL1	500	-	Back Pack securing screws	Check for secure seating.		W8645
QL1	500	2 a	Air filters	Fit new air filters.		W1005
QL1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL1	1000	2 a	Intermediate fuel filter	Replace intermediate fuel filter of filter element of intermediate fuel filter.		W1481
QL1	1000	2 a	Fuel prefilter	Fit new fuel prefilter or new fuel prefilter insert.		W1675
QL1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.	X	W1241
QL1	3000	1 a	Engine mounts	Carry out visual inspection of Trunnion mounts for general condition.		W1575
QL1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL1	3000	2 a	Exhaust gas recirculation	Check operation of exhaust flap actuators.		W1551
QL1	3000	3 a	Coolant expansion tank	Replace valve.	X	W1296
QL1	3000	18 a	Exhaust gas recirculation	Check coupling bars of EGR flaps for play.		W1817
QL1	4000	6 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL1	6000	6 a	NOx sensor	Fit new sensor.		W1525
QL1	6000	6 a	Oxygen sensor	Fit new sensor.		W1526
QL1	8000	18 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL1	8000	18 a	Turbochargers	Check turbine housing for cracks (externally by visual inspection and internally by endoscopic examination), replace if necessary..		W1577
QL1	8000	18 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL1	8000	18 a	Fuel injectors	Replace fuel injectors.		W1006
QL1	8000	18 a	Engine governor	Reset parameters of drift correction (CDC) and enter injector coding (IIG).		W1636
QL1	8000	18 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL1	12000	18 a	Cradle securing screws	Check for secure seating.		W4184
QL3	8000	18 a	Exhaust gas recirculation	Replace EGR actuators and check EGR flaps for wear, replace if necessary.		W1818
QL3	12000	6 a	Rubber sleeves	Replace all rubber sleeves.		W1250
w = weeks m = months a = years						

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL3	12000	6 a	Hose lines	Replace all hose lines.		W1251
QL3	12000	18 a	Component maintenance	Before starting maintenance work, carry out test run and record operating parameters. Then drain coolant and flush cooling systems.		W2012
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.		W2009
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110
QL3	12000	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	12000	18 a	Auxiliary PTO	Replace PTO.		W1735
QL3	12000	18 a	Turbochargers	Overhaul turbochargers.		W1038
QL3	24000	18 a	Battery charging alternator	Overhaul battery-charging generator.	X	W1043
QL3	24000	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL3	24000	18 a	HP fuel pump	Fit new high-pressure fuel sensor.		W1175
QL4	24000	18 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	24000	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
QL4	24000	18 a	Exhaust gas recirculation	Replace EGR cooler.		W1901
				Replace EGR flaps.		W1819
Transmission (ZF)						
QL1	Daily	-	Transmission operation	Check gear oil level.		X4032
				Check general condition of gearbox and inspect for leaks.		X4033
				Check for abnormal running noise and vibration.		X4034
QL1	500	-	Gear oil filter	Replace transmission oil filter.		X4035
QL1	500	1 a	Torsional resilient coupling	Check torsionally resilient coupling (visual inspection).		X4023
QL1	1000	-	Connectors/wiring harness	Check all electrical wiring components for chafing or damage.		X4037
QL4	24000	-	Transmission	Overhaul gearbox.		X4042
QL4	24000	-	Flexible/torsional resilient coupling	Replace rubber element.		W4146
w = weeks m = months a = years						

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Air compressor						
QL3	8000	-	Air compressor	Check air compressor and replace if required.	X	W1784
w = weeks m = months a = years						

1.5 Maintenance schedule matrix 21,000 h

0 to 14,000 Operating hours

Task	Limit	Operating hours [h]																														
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,250	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	
Engine																																
W0800	-																															
W0505	1 w																															
W1024	1 w																															
W1008	2 a																															
W1009	2 a																															
W0500	-	X																														
W0501	-	X																														
W0503	-	X																														
W0506	-	X																														
W0507	-	X																														
W0525	-	X																														
W8646	-	X																														
W8644	-	X																														
W8645	-		X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1005	2 a		X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
W1481	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
W1675	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
W1241	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
W1575	1 a							X						X						X							X					
W1207	2 a			X				X						X						X							X					
W1551	2 a							X						X						X							X					
W1296	3 a							X						X						X							X					
W1817	18 a							X						X						X							X					
W1046	6 a								X								X								X							X
W1525	6 a												X												X							
W1526	6 a												X												X							
W1610	18 a																X															X
W1577	18 a																X															X
W4184	18 a																									X						
W1689	18 a																									X						
W1006	18 a																									X						
W1636	18 a																									X						
W1011	18 a																									X						
W1818	18 a																X															X
W1250	6 a																									X						
W1251	6 a																									X						
W2012	18 a																									X						
W2001	18 a																									X						
W2006	18 a																									X						
W2007	18 a																									X						
W2009	18 a																									X						
w = weeks m = months a = years																																

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,250	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000
W2010	18 a																							X							
W2062	18 a																							X							
W2070	18 a																							X							
W2110	18 a																							X							
W1051	18 a																							X							
W1735	18 a																							X							
W1038	18 a																							X							
W1043	18 a																														
W1134	18 a																														
W1175	18 a																														
W3000	18 a																														
W3001	18 a																														
W3002	18 a																														
W3003	18 a																														
W3004	18 a																														
W3005	18 a																														
W3007	18 a																														
W3018	18 a																														
W3021	18 a																														
W3103	18 a																														
W3108	18 a																														
W3177	18 a																														
W3182	18 a																														
W1124	18 a																														
W1901	18 a																														
W1819	18 a																														
Transmission (ZF)																															
X4032	-	X																													
X4033	-	X																													
X4034	-	X																													
X4035	-		X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
X4023	1 a		X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
X4037	-			X		X		X		X		X			X		X		X		X		X		X		X		X		X
X4042	-																														
W4146	-																														
Air compressor																															
W1784	-																							X							
w = weeks m = months a = years																															

14,500 to 21,000 Operating hours

Task	Limit	Operating hours [h]																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
		14,500	15,000	15,500	15,750	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	

Task	Limit	Operating hours [h]																											
		14,500	15,000	15,500	15,750	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000													
W0505	1 w																												
W1024	1 w																												
W1008	2 a																												
W1009	2 a																												
W0500	-																												
W0501	-																												
W0503	-																												
W0506	-																												
W0507	-																												
W0525	-																												
W8646	-																												
W8644	-																												
W8645	-	X	X	X		X	X	X	X	X	X	X	X	X	X	X													
W1005	2 a	X	X	X		X	X	X	X	X	X	X	X	X	X	X													
W1001	2 a		X			X		X		X		X		X		X													
W1481	2 a		X			X		X		X		X		X		X													
W1675	2 a		X			X		X		X		X		X		X													
W1241	2 a		X			X		X		X		X		X		X													
W1575	1 a		X							X						X													
W1207	2 a		X							X						X													
W1551	2 a		X							X						X													
W1296	3 a		X							X						X													
W1817	18 a		X							X						X													
W1046	6 a								X							X													
W1525	6 a				X											X													
W1526	6 a				X											X													
W1610	18 a															X													
W1577	18 a															X													
W4184	18 a															X													
W1689	18 a															X													
W1006	18 a															X													
W1636	18 a															X													
W1011	18 a															X													
W1818	18 a															X													
W1250	6 a															X													
W1251	6 a															X													
W2012	18 a															X													
W2001	18 a															X													
W2006	18 a															X													
W2007	18 a															X													
W2009	18 a															X													
W2010	18 a															X													
W2062	18 a															X													
W2070	18 a															X													
W2110	18 a															X													
W1051	18 a															X													
W1735	18 a															X													

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																							
		14,500	15,000	15,500	15,750	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000									
W1038	18 a															X									
W1043	18 a															X									
W1134	18 a															X									
W1175	18 a															X									
W3000	18 a															X									
W3001	18 a															X									
W3002	18 a															X									
W3003	18 a															X									
W3004	18 a															X									
W3005	18 a															X									
W3007	18 a															X									
W3018	18 a															X									
W3021	18 a															X									
W3103	18 a															X									
W3108	18 a															X									
W3177	18 a															X									
W3182	18 a															X									
W1124	18 a															X									
W1901	18 a															X									
W1819	18 a															X									
Transmission (ZF)																									
X4032	-																								
X4033	-																								
X4034	-																								
X4035	-	X	X	X		X	X	X	X	X	X	X	X	X	X	X									
X4023	1 a	X	X	X		X	X	X	X	X	X	X	X	X	X	X									
X4037	-		X			X		X		X		X		X		X									
X4042	-															X									
W4146	-															X									
Air compressor																									
W1784	-															X									
w = weeks m = months a = years																									

1.6 Maintenance tasks 21,000 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL 1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL 1	-	1 w	Engine operation	Check relief bores of coolant pump(s).		W0505
QL 1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).	X	W1024
QL 1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL 1	-	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.		W1009
QL 1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.		W0503
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Drain off water and contamination from fuel prefilter.		W0507
				Inspect battery-charging generator for contamination and clean if necessary.	X	W0525
QL 1	Daily	-	Exhaust silencer	Check condensate drain of silencer.		W8646
QL 1	Daily	-	Back Pack securing screws	Check for damage (visual inspection).		W8644
QL 1	500	-	Back Pack securing screws	Check for secure seating.		W8645
QL 1	500	2 a	Air filters	Fit new air filters.		W1005
QL 1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL 1	1000	2 a	Intermediate fuel filter	Replace intermediate fuel filter of filter element of intermediate fuel filter.		W1481
QL 1	1000	2 a	Fuel prefilter	Fit new fuel prefilter or new fuel prefilter insert.		W1675
QL 1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.	X	W1241
QL 1	3000	1 a	Engine mounts	Carry out visual inspection of Trunnion mounts for general condition.		W1575
QL 1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL 1	3000	2 a	Exhaust gas recirculation	Check operation of exhaust flap actuators.		W1551
QL 1	3000	3 a	Coolant expansion tank	Replace valve.	X	W1296
QL 1	3000	18 a	Exhaust gas recirculation	Check coupling bars of EGR flaps for play.		W1817
QL 1	3500	6 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL 1	5250	6 a	NOx sensor	Fit new sensor.		W1525
QL 1	5250	6 a	Oxygen sensor	Fit new sensor.		W1526
QL 1	7000	18 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL 1	7000	18 a	Turbochargers	Check turbine housing for cracks (externally by visual inspection and internally by endoscopic examination), replace if necessary..		W1577
QL 1	10500	18 a	Cradle securing screws	Check for secure seating.		W4184
QL 1	10500	18 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL 1	10500	18 a	Fuel injectors	Replace fuel injectors.		W1006
QL 1	10500	18 a	Engine governor	Reset parameters of drift correction (CDC) and enter injector coding (IIG).		W1636
QL 1	10500	18 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL3	7000	18 a	Exhaust gas recirculation	Replace EGR actuators and check EGR flaps for wear, replace if necessary.		W1818
QL3	10500	6 a	Rubber sleeves	Replace all rubber sleeves.		W1250
w = weeks m = months a = years						

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL3	10500	6 a	Hose lines	Replace all hose lines.		W1251
QL3	10500	18 a	Component maintenance	Before starting maintenance work, carry out test run and record operating parameters. Then drain coolant and flush cooling systems.		W2012
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.		W2009
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110
QL3	10500	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	10500	18 a	Auxiliary PTO	Replace PTO.		W1735
QL3	10500	18 a	Turbochargers	Overhaul turbochargers.		W1038
QL3	21000	18 a	Battery charging alternator	Overhaul battery-charging generator.	X	W1043
QL3	21000	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL3	21000	18 a	HP fuel pump	Fit new high-pressure fuel sensor.		W1175
QL4	21000	18 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	21000	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
QL4	21000	18 a	Exhaust gas recirculation	Replace EGR cooler.		W1901
				Replace EGR flaps.		W1819
Transmission (ZF)						
QL1	Daily	-	Transmission operation	Check gear oil level.		X4032
				Check general condition of gearbox and inspect for leaks.		X4033
				Check for abnormal running noise and vibration.		X4034
QL1	500	-	Gear oil filter	Replace transmission oil filter.		X4035
QL1	500	1 a	Torsional resilient coupling	Check torsionally resilient coupling (visual inspection).		X4023
QL1	1000	-	Connectors/wiring harness	Check all electrical wiring components for chafing or damage.		X4037
QL4	21000	-	Transmission	Overhaul gearbox.		X4042
QL4	21000	-	Flexible/torsional resilient coupling	Replace rubber element.		W4146
w = weeks m = months a = years						

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Air compressor						
QL3	10500	-	Air compressor	Check air compressor and replace if required.	X	W1784
w = weeks m = months a = years						

1.7 Maintenance schedule matrix 18,000 h

0 to 13,000 Operating hours

Task	Limit	Operating hours [h]																														
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	3,600	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,200	7,500	8,000	8,500	9,000	9,500	10,000	10,500	10,800	11,000	11,500	12,000	12,500	13,000	
Engine																																
W0800	-																															
W0505	1 w																															
W1024	1 w																															
W1008	2 a																															
W1009	2 a																															
W0500	-	X																														
W0501	-	X																														
W0503	-	X																														
W0506	-	X																														
W0507	-	X																														
W0525	-	X																														
W8646	-	X																														
W8644	-	X																														
W8645	-		X	X	X	X	X	X	X		X	X	X	X	X	X	X		X	X	X	X	X	X	X		X	X	X	X	X	X
W1005	2 a		X	X	X	X	X	X	X		X	X	X	X	X	X	X		X	X	X	X	X	X	X		X	X	X	X	X	X
W1001	2 a			X		X		X			X		X		X		X			X		X		X			X		X		X	
W1481	2 a			X		X		X			X		X		X		X			X		X		X			X		X		X	
W1675	2 a			X		X		X			X		X		X		X			X		X		X			X		X		X	
W1241	2 a			X		X		X			X		X		X		X			X		X		X			X		X		X	
W1575	1 a							X							X							X							X			
W1207	2 a			X				X							X							X							X			
W1551	2 a							X							X							X							X			
W1296	3 a							X							X							X							X			
W1817	18 a							X							X							X							X			
W1046	6 a									X								X									X					
W1525	6 a														X														X			
W1526	6 a														X														X			
W1610	18 a																					X										
W4184	18 a																					X										
W1689	18 a																					X										
W1577	18 a																					X										
W1006	18 a																					X										
W1636	18 a																					X										
W1011	18 a																					X										
W1818	18 a														X														X			
W1250	6 a																					X										
W1251	6 a																					X										
W2012	18 a																					X										
W2001	18 a																					X										
W2006	18 a																					X										
W2007	18 a																					X										
W2009	18 a																					X										

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	3,600	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,200	7,500	8,000	8,500	9,000	9,500	10,000	10,500	10,800	11,000	11,500	12,000	12,500	13,000
W2010	18 a																					X									
W2062	18 a																					X									
W2070	18 a																					X									
W2110	18 a																					X									
W1051	18 a																					X									
W1735	18 a																					X									
W1038	18 a																					X									
W1043	18 a																														
W1134	18 a																														
W1175	18 a																														
W3000	18 a																														
W3001	18 a																														
W3002	18 a																														
W3003	18 a																														
W3004	18 a																														
W3005	18 a																														
W3007	18 a																														
W3018	18 a																														
W3021	18 a																														
W3103	18 a																														
W3108	18 a																														
W3177	18 a																														
W3182	18 a																														
W1124	18 a																														
W1901	18 a																														
W1819	18 a																														
Transmission (ZF)																															
X4032	-	X																													
X4033	-	X																													
X4034	-	X																													
X4035	-		X	X	X	X	X	X	X		X	X	X	X	X	X	X		X	X	X	X	X	X	X		X	X	X	X	X
X4023	1 a		X	X	X	X	X	X	X		X	X	X	X	X	X	X		X	X	X	X	X	X	X		X	X	X	X	X
X4037	-			X		X		X			X		X		X				X		X		X				X		X		X
X4042	-																														
W4146	-																														
Air compressor																															
W1784	-																					X									
w = weeks m = months a = years																															

13,500 to 18,000 Operating hours

Task	Limit	Operating hours [h]																	
		13,500	14,000	14,400	14,500	15,000	15,500	16,000	16,500	17,000	17,500	18,000							
Engine																			
W0800	-																		
w = weeks m = months a = years																			

Task	Limit	Operating hours [h]																											
		13,500	14,000	14,400	14,500	15,000	15,500	16,000	16,500	17,000	17,500	18,000																	
W0505	1 w																												
W1024	1 w																												
W1008	2 a																												
W1009	2 a																												
W0500	-																												
W0501	-																												
W0503	-																												
W0506	-																												
W0507	-																												
W0525	-																												
W8646	-																												
W8644	-																												
W8645	-	X	X		X	X	X	X	X	X	X	X																	
W1005	2 a	X	X		X	X	X	X	X	X	X	X																	
W1001	2 a		X			X		X		X		X																	
W1481	2 a		X			X		X		X		X																	
W1675	2 a		X			X		X		X		X																	
W1241	2 a		X			X		X		X		X																	
W1575	1 a					X						X																	
W1207	2 a					X						X																	
W1551	2 a					X						X																	
W1296	3 a					X						X																	
W1817	18 a					X						X																	
W1046	6 a			X								X																	
W1525	6 a											X																	
W1526	6 a											X																	
W1610	18 a											X																	
W4184	18 a											X																	
W1689	18 a											X																	
W1577	18 a											X																	
W1006	18 a											X																	
W1636	18 a											X																	
W1011	18 a											X																	
W1818	18 a											X																	
W1250	6 a											X																	
W1251	6 a					X						X																	
W2012	18 a											X																	
W2001	18 a											X																	
W2006	18 a											X																	
W2007	18 a											X																	
W2009	18 a											X																	
W2010	18 a											X																	
W2062	18 a											X																	
W2070	18 a											X																	
W2110	18 a											X																	
W1051	18 a											X																	
W1735	18 a											X																	

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																											
		13,500	14,000	14,400	14,500	15,000	15,500	16,000	16,500	17,000	17,500	18,000																	
W1038	18 a											X																	
W1043	18 a											X																	
W1134	18 a											X																	
W1175	18 a											X																	
W3000	18 a											X																	
W3001	18 a											X																	
W3002	18 a											X																	
W3003	18 a											X																	
W3004	18 a											X																	
W3005	18 a											X																	
W3007	18 a											X																	
W3018	18 a											X																	
W3021	18 a											X																	
W3103	18 a											X																	
W3108	18 a											X																	
W3177	18 a											X																	
W3182	18 a											X																	
W1124	18 a											X																	
W1901	18 a											X																	
W1819	18 a											X																	
Transmission (ZF)																													
X4032	-																												
X4033	-																												
X4034	-																												
X4035	-	X	X		X	X	X	X	X	X	X	X																	
X4023	1 a	X	X		X	X	X	X	X	X	X	X																	
X4037	-		X			X		X		X		X																	
X4042	-											X																	
W4146	-											X																	
Air compressor																													
W1784	-											X																	
w = weeks m = months a = years																													

1.8 Maintenance tasks 18,000 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL1	-	1 w	Engine operation	Check relief bores of coolant pump(s).		W0505
QL1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).	X	W1024
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL1	-	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.		W1009
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.		W0503
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Drain off water and contamination from fuel prefilter.		W0507
				Inspect battery-charging generator for contamination and clean if necessary.	X	W0525
QL1	Daily	-	Exhaust silencer	Check condensate drain of silencer.		W8646
QL1	Daily	-	Back Pack securing screws	Check for damage (visual inspection).		W8644
QL1	500	-	Back Pack securing screws	Check for secure seating.		W8645
QL1	500	2 a	Air filters	Fit new air filters.		W1005
QL1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL1	1000	2 a	Intermediate fuel filter	Replace intermediate fuel filter of filter element of intermediate fuel filter.		W1481
QL1	1000	2 a	Fuel prefilter	Fit new fuel prefilter or new fuel prefilter insert.		W1675
QL1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.	X	W1241
QL1	3000	1 a	Engine mounts	Carry out visual inspection of Trunnion mounts for general condition.		W1575
QL1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL1	3000	2 a	Exhaust gas recirculation	Check operation of exhaust flap actuators.		W1551
QL1	3000	3 a	Coolant expansion tank	Replace valve.	X	W1296
QL1	3000	18 a	Exhaust gas recirculation	Check coupling bars of EGR flaps for play.		W1817
QL1	3600	6 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL1	6000	6 a	NOx sensor	Fit new sensor.		W1525
QL1	6000	6 a	Oxygen sensor	Fit new sensor.		W1526
QL1	9000	18 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL1	9000	18 a	Cradle securing screws	Check for secure seating.		W4184
QL1	9000	18 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL1	9000	18 a	Turbochargers	Check turbine housing for cracks (externally by visual inspection and internally by endoscopic examination), replace if necessary..		W1577
QL1	9000	18 a	Fuel injectors	Replace fuel injectors.		W1006
QL1	9000	18 a	Engine governor	Reset parameters of drift correction (CDC) and enter injector coding (IIG).		W1636
QL1	9000	18 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL3	6000	18 a	Exhaust gas recirculation	Replace EGR actuators and check EGR flaps for wear, replace if necessary.		W1818
QL3	9000	6 a	Rubber sleeves	Replace all rubber sleeves.		W1250

w = weeks
m = months
a = years

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL3	9000	6 a	Hose lines	Replace all hose lines.		W1251
QL3	9000	18 a	Component maintenance	Before starting maintenance work, carry out test run and record operating parameters. Then drain coolant and flush cooling systems.		W2012
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.		W2009
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110
QL3	9000	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	9000	18 a	Auxiliary PTO	Replace PTO.		W1735
QL3	9000	18 a	Turbochargers	Overhaul turbochargers.		W1038
QL3	18000	18 a	Battery charging alternator	Overhaul battery-charging generator.	X	W1043
QL3	18000	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL3	18000	18 a	HP fuel pump	Fit new high-pressure fuel sensor.		W1175
QL4	18000	18 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	18000	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
QL4	18000	18 a	Exhaust gas recirculation	Replace EGR cooler.		W1901
				Replace EGR flaps.		W1819
Transmission (ZF)						
QL1	Daily	-	Transmission operation	Check gear oil level.		X4032
				Check general condition of gearbox and inspect for leaks.		X4033
				Check for abnormal running noise and vibration.		X4034
QL1	500	-	Gear oil filter	Replace transmission oil filter.		X4035
QL1	500	1 a	Torsional resilient coupling	Check torsionally resilient coupling (visual inspection).		X4023
QL1	1000	-	Connectors/wiring harness	Check all electrical wiring components for chafing or damage.		X4037
QL4	18000	-	Transmission	Overhaul gearbox.		X4042
QL4	18000	-	Flexible/torsional resilient coupling	Replace rubber element.		W4146
w = weeks m = months a = years						

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Air compressor						
QL3	9000	-	Air compressor	Check air compressor and replace if required.	X	W1784
w = weeks m = months a = years						

1.9 Maintenance schedule matrix 15,000 h

0 to 13,500 Operating hours

Task	Limit	Operating hours [h]																														
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	3,750	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,250	11,500	12,000	12,500	13,000	13,500	
Engine																																
W0800	-																															
W0505	1 w																															
W1024	1 w																															
W1008	2 a																															
W1009	2 a																															
W0500	-	X																														
W0501	-	X																														
W0503	-	X																														
W0506	-	X																														
W0507	-	X																														
W0525	-	X																														
W8646	-	X																														
W8644	-	X																														
W8645	-		X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X
W1005	2 a		X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X
W1001	2 a			X		X		X			X		X		X		X		X		X		X		X			X		X		
W1481	2 a			X		X		X			X		X		X		X		X		X		X		X			X		X		
W1675	2 a			X		X		X			X		X		X		X		X		X		X		X			X		X		
W1241	2 a			X		X		X			X		X		X		X		X		X		X		X			X		X		
W1575	1 a							X						X						X								X				
W1207	2 a			X				X						X						X								X				
W1551	2 a							X						X						X								X				
W1296	3 a							X						X						X								X				
W1817	18 a							X						X						X								X				
W1046	6 a								X								X										X					
W1525	6 a											X											X									
W1526	6 a											X											X									
W1610	18 a																	X														
W4184	18 a																	X														
W1689	18 a																	X														
W1577	18 a																	X														
W1006	18 a																	X														
W1636	18 a																	X														
W1011	18 a																	X														
W1818	18 a																	X														
W1250	6 a																															
W1251	6 a																															
W2012	18 a																															
W2001	18 a																															
W2006	18 a																															
W2007	18 a																															
W2009	18 a																															
w = weeks m = months a = years																																

w = weeks
m = months
a = years

[illegible]

14,000 to 15,000 Operating hours

[illegible]

Task	Limit	Operating hours [h]																											
		14,000	14,500	15,000																									
W0505	1 w																												
W1024	1 w																												
W1008	2 a																												
W1009	2 a																												
W0500	-																												
W0501	-																												
W0503	-																												
W0506	-																												
W0507	-																												
W0525	-																												
W8646	-																												
W8644	-																												
W8645	-	X	X	X																									
W1005	2 a	X	X	X																									
W1001	2 a	X		X																									
W1481	2 a	X		X																									
W1675	2 a	X		X																									
W1241	2 a	X		X																									
W1575	1 a			X																									
W1207	2 a			X																									
W1551	2 a			X																									
W1296	3 a			X																									
W1817	18 a			X																									
W1046	6 a			X																									
W1525	6 a			X																									
W1526	6 a			X																									
W1610	18 a			X																									
W4184	18 a			X																									
W1689	18 a			X																									
W1577	18 a			X																									
W1006	18 a			X																									
W1636	18 a			X																									
W1011	18 a			X																									
W1818	18 a			X																									
W1250	6 a			X																									
W1251	6 a			X																									
W2012	18 a			X																									
W2001	18 a			X																									
W2006	18 a			X																									
W2007	18 a			X																									
W2009	18 a			X																									
W2010	18 a			X																									
W2062	18 a			X																									
W2070	18 a			X																									
W2110	18 a			X																									
W1051	18 a			X																									
W1735	18 a			X																									

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																											
		14,000	14,500	15,000																									
W1038	18 a			X																									
W1043	18 a			X																									
W1134	18 a			X																									
W1175	18 a			X																									
W3000	18 a			X																									
W3001	18 a			X																									
W3002	18 a			X																									
W3003	18 a			X																									
W3004	18 a			X																									
W3005	18 a			X																									
W3007	18 a			X																									
W3018	18 a			X																									
W3021	18 a			X																									
W3103	18 a			X																									
W3108	18 a			X																									
W3177	18 a			X																									
W3182	18 a			X																									
W1124	18 a			X																									
W1901	18 a			X																									
W1819	18 a			X																									
Transmission (ZF)																													
X4032	-																												
X4033	-																												
X4034	-																												
X4035	-	X	X	X																									
X4023	1 a	X	X	X																									
X4037	-	X		X																									
X4042	-			X																									
W4146	-			X																									
Air compressor																													
W1784	-			X																									
w = weeks m = months a = years																													

1.10 Maintenance tasks 15,000 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL 1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL 1	-	1 w	Engine operation	Check relief bores of coolant pump(s).		W0505
QL 1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).	X	W1024
QL 1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL 1	-	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.		W1009
QL 1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.		W0503
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Drain off water and contamination from fuel prefilter.		W0507
				Inspect battery-charging generator for contamination and clean if necessary.	X	W0525
QL 1	Daily	-	Exhaust silencer	Check condensate drain of silencer.		W8646
QL 1	Daily	-	Back Pack securing screws	Check for damage (visual inspection).		W8644
QL 1	500	-	Back Pack securing screws	Check for secure seating.		W8645
QL 1	500	2 a	Air filters	Fit new air filters.		W1005
QL 1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL 1	1000	2 a	Intermediate fuel filter	Replace intermediate fuel filter of filter element of intermediate fuel filter.		W1481
QL 1	1000	2 a	Fuel prefilter	Fit new fuel prefilter or new fuel prefilter insert.		W1675
QL 1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.	X	W1241
QL 1	3000	1 a	Engine mounts	Carry out visual inspection of Trunnion mounts for general condition.		W1575
QL 1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL 1	3000	2 a	Exhaust gas recirculation	Check operation of exhaust flap actuators.		W1551
QL 1	3000	3 a	Coolant expansion tank	Replace valve.	X	W1296
QL 1	3000	18 a	Exhaust gas recirculation	Check coupling bars of EGR flaps for play.		W1817
QL 1	3750	6 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL 1	5000	6 a	NOx sensor	Fit new sensor.		W1525
QL 1	5000	6 a	Oxygen sensor	Fit new sensor.		W1526
QL 1	7500	18 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL 1	7500	18 a	Cradle securing screws	Check for secure seating.		W4184
QL 1	7500	18 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL 1	7500	18 a	Turbochargers	Check turbine housing for cracks (externally by visual inspection and internally by endoscopic examination), replace if necessary..		W1577
QL 1	7500	18 a	Fuel injectors	Replace fuel injectors.		W1006
QL 1	7500	18 a	Engine governor	Reset parameters of drift correction (CDC) and enter injector coding (IIG).		W1636
QL 1	7500	18 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL3	7500	18 a	Exhaust gas recirculation	Replace EGR actuators and check EGR flaps for wear, replace if necessary.		W1818
QL3	15000	6 a	Rubber sleeves	Replace all rubber sleeves.		W1250
w = weeks m = months a = years						

TIM-ID: 0000063230 - 003

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL3	15000	6 a	Hose lines	Replace all hose lines.		W1251
QL3	15000	18 a	Component maintenance	Before starting maintenance work, carry out test run and record operating parameters. Then drain coolant and flush cooling systems.		W2012
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.		W2009
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110
QL3	15000	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	15000	18 a	Auxiliary PTO	Replace PTO.		W1735
QL3	15000	18 a	Turbochargers	Overhaul turbochargers.		W1038
QL3	15000	18 a	Battery charging alternator	Overhaul battery-charging generator.	X	W1043
QL3	15000	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL3	15000	18 a	HP fuel pump	Fit new high-pressure fuel sensor.		W1175
QL4	15000	18 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	15000	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
QL4	15000	18 a	Exhaust gas recirculation	Replace EGR cooler.		W1901
				Replace EGR flaps.		W1819
Transmission (ZF)						
QL1	Daily	-	Transmission operation	Check gear oil level.		X4032
				Check general condition of gearbox and inspect for leaks.		X4033
				Check for abnormal running noise and vibration.		X4034
QL1	500	-	Gear oil filter	Replace transmission oil filter.		X4035
QL1	500	1 a	Torsional resilient coupling	Check torsionally resilient coupling (visual inspection).		X4023
QL1	1000	-	Connectors/wiring harness	Check all electrical wiring components for chafing or damage.		X4037
QL4	15000	-	Transmission	Overhaul gearbox.		X4042
QL4	15000	-	Flexible/torsional resilient coupling	Replace rubber element.		W4146
w = weeks m = months a = years						

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Air compressor						
QL3	7500	-	Air compressor	Check air compressor and replace if required.	X	W1784
w = weeks m = months a = years						

1.11 Maintenance schedule matrix 12,000 h

0 to 12,000 Operating hours

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000					
Engine																															
W0800	-																														
W0505	1 w																														
W1024	1 w																														
W1008	2 a																														
W1009	2 a																														
W0500	-	X																													
W0501	-	X																													
W0503	-	X																													
W0506	-	X																													
W0507	-	X																													
W0525	-	X																													
W8646	-	X																													
W8644	-	X																													
W8645	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1005	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1481	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1675	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a							X						X						X								X			
W1207	2 a			X				X						X						X								X			
W1551	2 a							X						X						X								X			
W1296	3 a							X						X						X								X			
W1817	18 a							X						X						X								X			
W1046	6 a									X								X										X			
W1525	6 a													X														X			
W1526	6 a													X														X			
W1610	18 a													X														X			
W4184	18 a													X														X			
W1689	18 a													X														X			
W1577	18 a													X														X			
W1006	18 a													X														X			
W1636	18 a													X														X			
W1011	18 a													X														X			
W1818	18 a													X														X			
W1250	6 a																											X			
W1251	6 a																											X			
W2012	18 a																											X			
W2001	18 a																											X			
W2006	18 a																											X			
W2007	18 a																											X			
W2009	18 a																											X			

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																														
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000						
W2010	18 a																										X					
W2062	18 a																										X					
W2070	18 a																										X					
W2110	18 a																										X					
W1051	18 a																										X					
W1735	18 a																										X					
W1038	18 a																										X					
W1043	18 a																										X					
W1134	18 a																										X					
W1175	18 a																										X					
W3000	18 a																										X					
W3001	18 a																										X					
W3002	18 a																										X					
W3003	18 a																										X					
W3004	18 a																										X					
W3005	18 a																										X					
W3007	18 a																										X					
W3018	18 a																										X					
W3021	18 a																										X					
W3103	18 a																										X					
W3108	18 a																										X					
W3177	18 a																										X					
W3182	18 a																										X					
W1124	18 a																										X					
W1901	18 a																										X					
W1819	18 a																										X					
Transmission (ZF)																																
X4032	-	X																														
X4033	-	X																														
X4034	-	X																														
X4035	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
X4023	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
X4037	-			X		X		X		X		X		X		X		X		X		X		X		X		X				
X4042	-																															
W4146	-																															
Air compressor																																
W1784	-												X														X					
w = weeks m = months a = years																																

1.12 Maintenance tasks 12,000 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL1	-	1 w	Engine operation	Check relief bores of coolant pump(s).		W0505
QL1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).	X	W1024
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL1	-	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.		W1009
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.		W0503
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Drain off water and contamination from fuel prefilter.		W0507
				Inspect battery-charging generator for contamination and clean if necessary.	X	W0525
QL1	Daily	-	Exhaust silencer	Check condensate drain of silencer.		W8646
QL1	Daily	-	Back Pack securing screws	Check for damage (visual inspection).		W8644
QL1	500	-	Back Pack securing screws	Check for secure seating.		W8645
QL1	500	2 a	Air filters	Fit new air filters.		W1005
QL1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL1	1000	2 a	Intermediate fuel filter	Replace intermediate fuel filter of filter element of intermediate fuel filter.		W1481
QL1	1000	2 a	Fuel prefilter	Fit new fuel prefilter or new fuel prefilter insert.		W1675
QL1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.	X	W1241
QL1	3000	1 a	Engine mounts	Carry out visual inspection of Trunnion mounts for general condition.		W1575
QL1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL1	3000	2 a	Exhaust gas recirculation	Check operation of exhaust flap actuators.		W1551
QL1	3000	3 a	Coolant expansion tank	Replace valve.	X	W1296
QL1	3000	18 a	Exhaust gas recirculation	Check coupling bars of EGR flaps for play.		W1817
QL1	4000	6 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL1	6000	6 a	NOx sensor	Fit new sensor.		W1525
QL1	6000	6 a	Oxygen sensor	Fit new sensor.		W1526
QL1	6000	18 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL1	6000	18 a	Cradle securing screws	Check for secure seating.		W4184
QL1	6000	18 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL1	6000	18 a	Turbochargers	Check turbine housing for cracks (externally by visual inspection and internally by endoscopic examination), replace if necessary..		W1577
QL1	6000	18 a	Fuel injectors	Replace fuel injectors.		W1006
QL1	6000	18 a	Engine governor	Reset parameters of drift correction (CDC) and enter injector coding (IIG).		W1636
QL1	6000	18 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL3	6000	18 a	Exhaust gas recirculation	Replace EGR actuators and check EGR flaps for wear, replace if necessary.		W1818
QL3	12000	6 a	Rubber sleeves	Replace all rubber sleeves.		W1250

w = weeks
m = months
a = years

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL3	12000	6 a	Hose lines	Replace all hose lines.		W1251
QL3	12000	18 a	Component maintenance	Before starting maintenance work, carry out test run and record operating parameters. Then drain coolant and flush cooling systems.		W2012
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.		W2009
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110
QL3	12000	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	12000	18 a	Auxiliary PTO	Replace PTO.		W1735
QL3	12000	18 a	Turbochargers	Overhaul turbochargers.		W1038
QL3	12000	18 a	Battery charging alternator	Overhaul battery-charging generator.	X	W1043
QL3	12000	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL3	12000	18 a	HP fuel pump	Fit new high-pressure fuel sensor.		W1175
QL4	12000	18 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	12000	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
QL4	12000	18 a	Exhaust gas recirculation	Replace EGR cooler.		W1901
				Replace EGR flaps.		W1819
Transmission (ZF)						
QL1	Daily	-	Transmission operation	Check gear oil level.		X4032
				Check general condition of gearbox and inspect for leaks.		X4033
				Check for abnormal running noise and vibration.		X4034
QL1	500	-	Gear oil filter	Replace transmission oil filter.		X4035
QL1	500	1 a	Torsional resilient coupling	Check torsionally resilient coupling (visual inspection).		X4023
QL1	1000	-	Connectors/wiring harness	Check all electrical wiring components for chafing or damage.		X4037
QL4	12000	-	Transmission	Overhaul gearbox.		X4042
QL4	12000	-	Flexible/torsional resilient coupling	Replace rubber element.		W4146
w = weeks m = months a = years						

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Air compressor						
QL3	6000	-	Air compressor	Check air compressor and replace if required.	X	W1784
w = weeks m = months a = years						